

PHY101: Assignment - 1

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Problem 1

A coin (assumed to be a point mass) lies on the top of a card which, in turn, is placed over the rim of a drinking glass of radius a . The card just covers the glass and initially the coin is placed just above the center of the glass. With what minimum velocity, assumed to be uniform, should the card be snapped so that the coin falls into the glass. Coefficient of friction between the card and coin is μ and that between the card and glass is negligible.

Hint: $v^2 > 2\mu ga$.

Problem 2

A boat of mass m is pushed and imparted an initial velocity v_0 . Assuming the force due to water resistance to be μv , determine the time in which the velocity will drop to half of its initial value and the distance the boat will travel during this time. Determine also the distance the boat will travel before it stops.

Problem 3

A particle of mass m is traveling along the x -axis such that at $t = 0$ it is located at $x = 0$ and has an initial speed v_0 . The particle is acted upon by an opposing force βv^2 , v being the speed of the particle. Find speed, position and acceleration of the particle at any subsequent time. Show that the particle will never come to rest.

Problem 4

A body of mass m is projected vertically upward in a medium in which it experiences a resistance of mk^2v^2 . If the initial velocity is v_0 , show that the body returns to the point of projection with a velocity v_1 such that

$$v_1^2 = \frac{gv_0^2}{g+k^2v_0^2}.$$

Hint: Find the maximum height attained by the particle and check your result in the limit $k \rightarrow 0$.

Problem 5

A uniform rope of mass m and length l is hung vertically from a pivot. Find an expression for the tension at any point of the rope.

Hint: Write down the equation of balance of force for an infinitesimal section of the rope and use the boundary condition $T(l) = 0$.